

CHAPTER 18

SOILS AND FOUNDATIONS

SECTION BC 1801 GENERAL AND DEFINITIONS

1801.1 Scope. The provisions of this chapter shall apply to building and foundation systems in those areas not subject to scour or water pressure by wind and wave action. Buildings and foundations subject to such scour or water pressure loads shall be designed in accordance with Chapter 16 and Appendix G. *(Note: Where the text in this Code refers to ASCE 7, the 2005 edition shall be used; and where the text in this Code refers to ASCE 7-10, the 2010 edition shall be used.)*

1801.2 Design. Allowable bearing pressures, allowable stresses and design formulas provided in this chapter shall be used with the allowable stress design load combinations specified in Section 1605.3. The quality and design of materials used structurally in excavations and foundations shall conform to the requirements specified in Chapters 16, 19, 21, 22 and 23. Excavations and fills shall also comply with Chapter 33.

1801.2.1 Foundation design for seismic overturning. Where foundations are proportioned using the load combinations of Sections 1605.2 or 1605.3.1 and the computation of seismic overturning effects is by equivalent lateral force or modal analysis, the proportioning shall be in accordance with Section 12.13.4 of ASCE 7-10.

1801.3 Definitions. The following terms shall, for the purposes of this chapter, have the meanings shown herein.

AUGERED-CAST-IN-PLACE PILES. Augered-cast-in-place piles are constructed by pumping grout into an augered hole during the withdrawal of the auger. The pile is reinforced with a single reinforcing bar, a reinforcing steel cage or a structural steel section.

CAISSON PILES. Steel cased piles constructed by advancing a steel shell seated into rock and drilling of an uncased socket within the rock. The shell and socket are filled with a steel core section or steel reinforcing, and concrete or grout.

COMPACTED CONCRETE PILES. Compacted concrete piles are constructed by filling a driven casing with low-strength concrete and compacting the concrete as the casing is withdrawn.

COMPOSITE PILES. Composite piles consist of two or more approved pile types joined together.

CONCRETE-FILLED STEEL PIPE AND TUBE PILES. Concrete-filled steel pipe and tube piles are constructed by driving a steel pipe or tube section into the soil and filling the pipe or tube section with concrete. The steel pipe or tube section is left in place during and after the deposition of the concrete. For the purposes of this code these piles shall be considered driven piles.

DAMPPROOFING. Dampproofing is a protective measure applied to building foundation walls and slabs to prevent moisture from passing into interior spaces.

DEEP FOUNDATIONS. Deep foundations are comprised of concrete, grout, wood or steel structural elements either driven, drilled or jacked into the ground or cast in place. Deep foundations are relatively slender in comparison to their length, with lengths exceeding 12 times the least horizontal dimension. Deep foundations derive their load-carrying capacity through skin friction, end bearing, or a combination thereof.

DRIVEN UNCASED PILES. Driven uncased piles are constructed by driving a steel shell into the soil to shore an unexcavated hole that is later filled with concrete. The steel casing is lifted out of the hole during the deposition of the concrete. Driven uncased piles are not permitted under the provisions of this code.

ENLARGED BASE PILES. Enlarged base piles are cast-in-place concrete piles constructed with a base that is larger than the diameter of the remainder of the pile. The enlarged base is designed to increase the load-bearing area of the pile in end bearing. Enlarged base piles include piles installed by driving a precast concrete tip or by compacting concrete into the base of the pile to form an enlarged base.

HELICAL PILES. Helical piles are manufactured deep foundation steel elements consisting of a shaft and one or more helical bearing plates (helices) screwed into the ground by application of torque on the shaft. The various products marketed as screw piles, torque anchors, and helical piers are considered helical piles.

H-PILES. Steel H-piles are constructed by driving a steel H-shaped section into the ground.

FIXED HEADED PILE (DEEP FOUNDATION). A pile connected to a pile cap in a manner that prevents rotation

of the pile head.

FREE HEADED PILE. A pile with a head that is free to rotate.

GEOTECHNICAL CAPACITY OF DEEP FOUNDATIONS. The load that can be supported by the soil or rock surrounding deep foundation as determined using a recognized method of analysis or as established by load tests. The geotechnical capacity can be developed through skin friction, end bearing, or a combination thereof.

LIQUEFACTION. For granular soils, liquefaction is defined as the loss of shear strength in soils resulting from increased pore-water pressure and reduced effective stress that may develop as a result of cyclic loading during earthquakes. For cohesive soils with a plasticity index of less than 20, liquefaction is defined as any transient softening and increased cyclic shear strains that may occur during earthquakes.

MICROPILE. A micropile is a drilled and grouted deep foundation element with a diameter that measures 5-inches (127 mm) to 14-inches (356 mm) that develops its load-carrying capacity by means of a bond zone in soil (also commonly known as a minipile).

PIER FOUNDATION. A pier foundation is a shallow foundation element of masonry or cast-in-place concrete construction. Piers are relatively short in comparison to their width, with lengths less than or equal to 12 times the least horizontal dimension of the pier. Piers derive their load-carrying capacity from end bearing on soil or rock.

RETAINING WALL. A wall that resists lateral or other forces caused by soil, rock, water or other materials, thereby limiting lateral displacement and the movement of the supported materials. Basement walls and vault walls that are parts of buildings and underground structures, including but not limited to utility vault structures, tunnels and transit stations, are not considered retaining walls.

SHALLOW FOUNDATION. A shallow foundation is an individual or strip footing, a mat foundation, a slab-on-grade foundation or a similar foundation element.

UNDERPINNING. The alteration of an existing foundation to transfer loads to a lower bearing stratum using new piers, piles, or other structural support elements installed below the existing foundation.

WATERPROOFING. Waterproofing is a protective measure applied to building foundation walls and slabs to prevent moisture and liquid water from passing into interior spaces.

SECTION BC 1802

GEOTECHNICAL INVESTIGATIONS AND MATERIAL CLASSIFICATIONS

1802.1 General. Geotechnical investigations shall be subject to special inspections in accordance with Sections 1704.7, 1704.8 and 1704.9 and be conducted in conformance with Sections 1802.2 through 1802.7. An engineer shall scope, supervise and approve the subsurface investigation and the classification of the soil and rock encountered.

1802.2 Where required. A geotechnical investigation shall be conducted for:

1. New structures;
2. Horizontal enlargements;
3. Vertical enlargements or alterations necessitating new foundations or resulting in additional loading that exceeds 5 percent of the existing foundation design capacity; or
4. As required by the commissioner or applicant of record.

The geotechnical investigation shall be performed in accordance with Sections 1802.4 through 1802.6. For structures in Seismic Design Category C or D, the requirements of Sections 1802.2.1 and 1802.2.2 shall also apply.

1802.2.1 Seismic Design Category C. Where a structure is determined to be in Seismic Design Category C in accordance with Section 1613, the geotechnical investigation shall include an evaluation of the following potential hazards resulting from earthquake motions: slope instability, liquefaction and surface rupture due to faulting or lateral spreading. Peak ground acceleration for use in liquefaction analyses shall be determined in accordance with Section 1813.2.1.

1802.2.2 Seismic Design Category D. Where a structure is determined to be in Seismic Design Category D in accordance with Section 1613, the requirements for Seismic Design Category C, given in Section 1802.2.1, shall be met. In addition, the following shall be conducted:

1. A determination of lateral pressures on basement, cellar, and retaining walls due to earthquake motions. Peak ground acceleration for use in lateral pressure analyses shall be determined in accordance with Section 1813.2.1.

2. An assessment of potential consequences of any liquefaction and soil strength loss, including estimation of differential settlement, lateral movement or reduction in foundation soil-bearing capacity. Mitigation measures shall be addressed. Such measures shall be given consideration in the design of the structure and shall include, but are not limited to, ground stabilization, selection of appropriate foundation type and depths, selection of appropriate structural systems to accommodate anticipated displacements or any combination of these measures.

1802.3 Material classification. Soil and rock classification shall be based on materials disclosed by borings, test pits or other subsurface exploration methods. Soil classifications shall be determined in accordance with ASTM D 2487 (refer to Table 1802.3) and the supplemental definitions contained herein. Rock classifications shall be determined in accordance with generally accepted engineering practice and the supplemental definitions contained herein. Laboratory tests shall be conducted to ascertain these classifications where deemed necessary by the engineer responsible for the geotechnical investigation or the commissioner.

BEDROCK.

1. **Hard sound rock (Class 1a).** Includes crystalline rocks, such as gneiss, granite, diabase and mica schist. Characteristics are as follows: the rock rings when struck with pick or bar; the rock does not disintegrate after exposure to air or water; the rock breaks with sharp fresh fracture; cracks are unweathered, less than 1/8-inch (3.2 mm) wide, and generally no closer than 3 feet (914 mm) apart; and the RQD (rock quality designation) with a double tube, NX-size diamond core barrel is generally 85 percent or greater for each 5-foot (1524 mm) run; or core recovery with BX-size core is generally 85 percent or greater for each 5-foot (1524 mm) run.
2. **Medium hard rock (Class 1b).** Includes crystalline rocks of paragraph (1) of this subdivision, plus marble and serpentinite. Characteristics are as follows: all those listed in paragraph (1) of this subdivision, except that cracks may be 1/4-inch (6.4 mm) wide and slightly weathered, generally spaced no closer than 2 feet (610 mm) apart; and the RQD with a double tube, NX-size diamond core barrel is generally between 50 and 85 percent for each 5-foot (1524 mm) run; or core recovery with BX-size core is generally 50 to 85 percent for each 5-foot (1524 mm) run.
3. **Intermediate rock (Class 1c).** Includes rocks described in paragraphs (1) and (2) of this subdivision, plus cemented shales and sandstone. Characteristics are as follows: the rock gives dull sound when struck with pick or bar; does not disintegrate after exposure to air or water; broken pieces may show moderately weathered surfaces; may contain fracture and moderately weathered zones up to 1 inch (25 mm) wide spaced as close as 1 foot (305 mm) apart; and the RQD with a double tube, NX-size diamond core barrel is generally 35 to 50 percent for each 5-foot (1524 mm) run; or a core recovery with BX-size core of generally 35 to 50 percent for each 5-foot (1524 mm) run.
4. **Soft rock (Class 1d).** Includes rocks described in paragraphs (1), (2), and (3) of this subdivision in highly weathered condition, plus talc schist and poorly cemented shales and sandstones. Characteristics are: rock may soften on exposure to air or water; may contain highly weathered zones up to 3 inches (76 mm) wide but filled with stiff soil; and either the RQD with a double tube, NX-size diamond core barrel is less than 35 percent for each 5-foot (1524 mm) run or core recovery with BX-size core of generally less than 35 percent for each 5-foot (1524 mm) run, or a standard penetration resistance more than 50 blows per foot (0.3 meters).

SANDY GRAVEL AND GRAVELS. Consists of coarse-grained material with more than half of the coarse fraction larger than the #4 size sieve and contains little or no fines (GW and GP). The density of these materials shall be determined in accordance with the following:

Dense (Class 2a). These materials have a standard penetration test N-value greater than 30 blows per 1 foot (0.3 meter).

Medium (Class 2b). These materials have a standard penetration test N-value between 10 and 30 blows per 1 foot (0.3 meter).

Loose (Class 6). These materials have a standard penetration test N-value less than 10 blows per 1 foot (0.3 meter). These materials shall be considered nominally unsatisfactory bearing materials.

GRANULAR SOILS. These materials are coarse-grained soils consisting of gravel and/or sand with appreciable amounts of fines, and gravel. Soil types include GM, GC, SW, SP, SM, and SC. The density of granular materials shall be determined in accordance with the following:

Dense (Class 3a). These materials have a standard penetration test N-value of greater than 30 blows per 1 foot (0.3 meter).

Medium (Class 3b). These materials have a standard penetration test N-value of between 10 and 30 blows per 1

foot (0.3 meter).

Loose (Class 6). These materials have standard penetration test N-value of fewer than 10 blows per 1 foot (0.3 meter). These materials shall be considered nominally unsatisfactory bearing materials.

CLAYS. For soil types SC, CL and CH in the absence of sufficient laboratory data, the consistency of clay materials shall be determined in accordance with the following:

Hard (Class 4a). Clay requiring picking for removal, a fresh sample of which cannot be molded by pressure of the fingers; or having an unconfined compressive strength in excess of 4 TSF (383 kPa); or having a standard penetration test where the N-value is greater than 30 blows per 1 foot (0.3 meter).

Stiff (Class 4b). Clay that can be removed by spading, a fresh sample of which requires substantial pressure of the fingers to create an indentation; or having an unconfined compressive strength of between 1 TSF (96 kPa) and 4 TSF (383 kPa); or having a standard penetration test where the N-value is between 8 and 30 blows per 1 foot (0.3 meter).

Medium (Class 4c). Clay that can be removed by spading, a fresh sample of which can be molded by substantial pressure of the fingers; or having an unconfined compressive strength of between 0.5 TSF (48 kPa) and 1 TSF (96 kPa); or having a standard penetration test where the N-value is between 4 and 8 blows per 1 foot (0.3 meter).

Soft (Class 6). Clay, a fresh sample of which can be molded with slight pressure of the fingers; or having an unconfined compressive strength of less than 0.5 TSF (48 kPa); or having a standard penetration test where the N-value is fewer than 4 blows per 1 foot (0.3 meter). This material shall be considered nominally unsatisfactory bearing material.

SILTS AND CLAYEY SILTS. For soil types ML and MH in the absence of sufficient laboratory data, the consistency of silt materials shall be determined in accordance with the following:

Dense (Class 5a). Silt with a standard penetration test where the N-value is greater than 30 blows per 1 foot (0.3 meter).

Medium (Class 5b). Silt with a standard penetration test where the N-value is between 10 and 30 blows per 1 foot (0.3 meter).

Loose (Class 6). Silt with a standard penetration test where the N-value is fewer than 10 blows per 1 foot (0.3 meters). This material shall be considered nominally unsatisfactory bearing material.